



EQUITY RESEARCH | INDIRA SECURITIES

The Indian Battery Industry

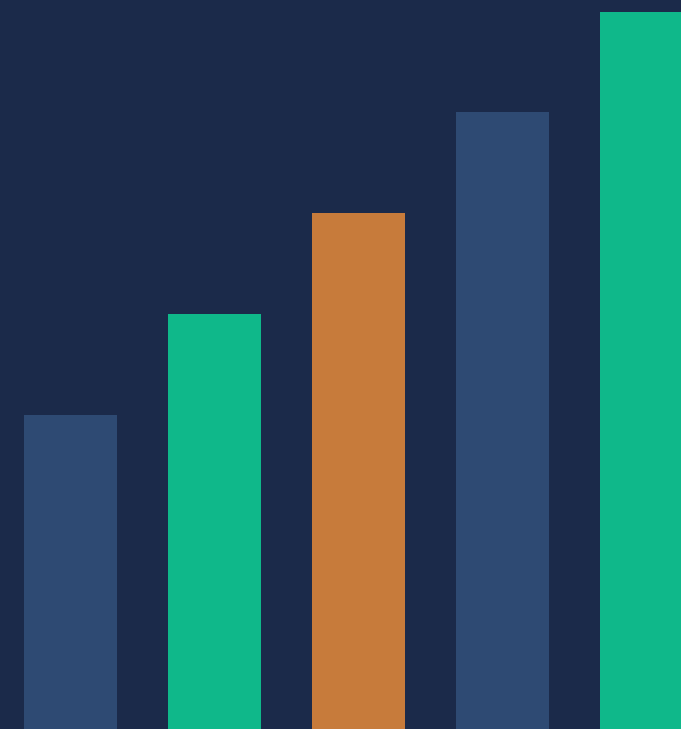
From Lead-Acid to Lithium-Ion

A Coverage Initiation Framework on Exide Industries & Amara Raja Energy & Mobility

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How This Presentation Flows

01

The Technology

What lead-acid and lithium-ion batteries are, how they differ, and why it matters

02

Raw Materials & Supply Chain

Lead, lithium, nickel, cobalt, graphite — and India's import dependence

03

The Indian Industry

Market structure, growth drivers, government policy, the gigafactory race

04

Exide Industries

Business model, new-energy pivot (EESL), financials, investment thesis

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Amara Raja Energy & Mobility

Business model, Giga Corridor, financials, investment thesis

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Head-to-Head & Valuation

Comparative scorecard, SOTP valuation, risks, and monitoring framework

Two Incumbents, One Parallel Bet

Exide and Amara Raja dominate India's mature lead-acid duopoly. Both are now using that cash-generative core to fund an expensive, high-risk pivot into lithium-ion cell manufacturing — a technology neither company has an incumbency advantage in.

165+

years lead-acid has been in continuous commercial production

~2 GWh

India's current operational lithium-ion capacity

180–220+

GWh of announced domestic gigafactory pipeline by 2030

2

listed companies at the centre of this dossier: Exide & Amara Raja

This presentation builds the full picture from first principles — the chemistry, the supply chain, the industry — before arriving at a company-level view and a valuation.



PART ONE

Understanding the Technology

What a battery actually is, how lead-acid and lithium-ion differ, and why that difference is reshaping a 165-year-old industry.

What a Battery Actually Does

A battery stores energy chemically and releases it as electric current through a reaction between two electrodes.

1

Anode

The negative electrode. Gives up electrons (oxidised) during discharge.

2

Cathode

The positive electrode. Accepts electrons (reduced) during discharge.

3

Electrolyte

Lets ions — not electrons — move between electrodes, forcing current through the external circuit.

4

Separator

A porous membrane that keeps electrodes apart while letting ions pass.



A secondary (rechargeable) battery — the only kind that matters here — reverses this reaction on charge. Every chemistry in this deck (lead-acid, lithium-ion, sodium-ion) is rechargeable.

Lead-Acid vs. Lithium-Ion

	LEAD-ACID	LITHIUM-ION
Invented	1859 (Gaston Planté)	1991 (Sony, commercial)
Energy density	30–50 Wh/kg	150–250+ Wh/kg
Cycle life	300–500 cycles	1,000–6,000+ cycles
Relative cost	Low	High (falling)
Safety profile	Robust, well understood	Needs active BMS management
Primary use today	Auto SLI, UPS, telecom, data-centre backup	EVs, BESS, telecom/data-centre (migrating)



Neither wins outright

Lithium-ion wins on weight and lifespan; lead-acid wins on cost, safety margin and near-100% recyclability. That’s why both will coexist for years.

LFP vs. NMC — The Chemistry Choice That Shapes Strategy

Within lithium-ion, the single biggest strategic decision every Indian manufacturer is making right now is how to split capacity between two cathode chemistries.

LFP

Lithium Iron Phosphate

- Moderate energy density (~120–160 Wh/kg)
- Very high cycle life: 3,000–6,000+ cycles
- Excellent thermal safety — no nickel/cobalt
- Lowest cost per kWh
- Best for: BESS, standard-range EVs, telecom, 3-wheelers

~60–65% of India's 2030s lithium-ion demand

NMC

Nickel Manganese Cobalt

- High energy density (~200–250+ Wh/kg)
- Moderate cycle life: 1,000–2,000 cycles
- Higher thermal-runaway risk than LFP
- Higher, more volatile cost (Ni + Co)
- Best for: long-range passenger EVs, premium 2-wheelers

~25–30% of India's 2030s lithium-ion demand



PART TWO

Raw Materials & Supply Chain

Lead-acid runs on a short, circular, largely domestic supply chain. Lithium-ion runs on a long chain India barely controls today.

Two Very Different Value Chains

LEAD-ACID — SHORT & CIRCULAR



95–99% of lead in a spent battery is recoverable — Exide (Chloride Metals) and Amara Raja (ARCS) both recycle at scale, cutting import dependence.

LITHIUM-ION — LONG & IMPORT-DEPENDENT



Navy = almost entirely imported into India today. Teal = where India's domestic capability is beginning to build (cell manufacturing and downstream).

Bottom line: Exide and Amara Raja both know how to run the lead-acid chain end-to-end. In lithium-ion, they are building brand-new capability at the one stage (cell manufacturing) where India has the least experience — alongside Reliance, Tata, Ola and others.

Where the Inputs Come From

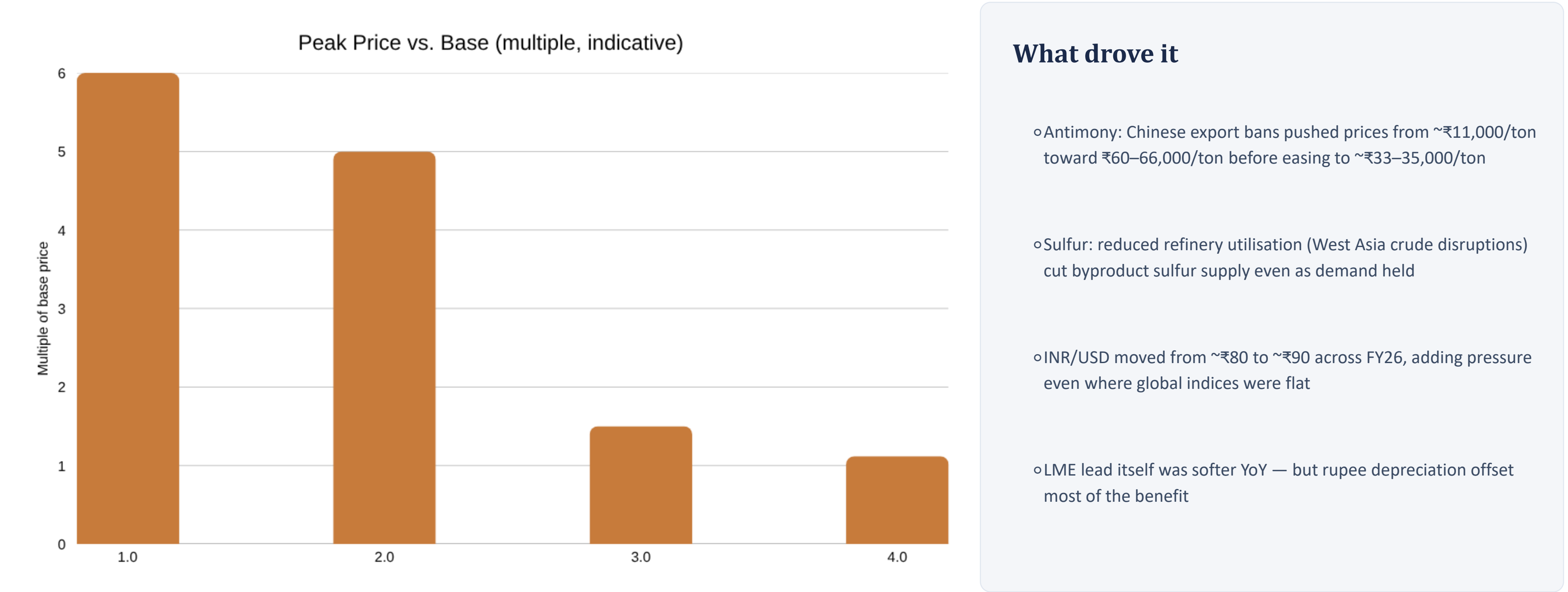
Material	Dominant Global Source	India's Position
Lead	China, Australia, Peru, USA, Mexico	Partial — offset by strong domestic recycling
Lithium	Chile, Australia, Argentina, China	Near-total; first J&K discovery (5.9 Mt) is 5–7+ yrs from production
Nickel	Indonesia (dominant), Australia, Brazil	Fully import-dependent
Cobalt	DR Congo (~70% of mined output)	Fully import-dependent; highest ESG-risk input
Graphite	China (70%+ of processed supply)	Import-dependent, especially battery-grade grades



Why this matters for the model: China dominates refining for nearly every lithium-ion input and periodically tightens export controls on raw materials, precursor chemicals, and technology transfer — a real, dated risk (a China export-restriction delay to November 2026 is already on the calendar) for both companies’ Chinese technology partners.

Commodity Volatility Is Squeezing Both Companies Today

FY26 saw extraordinary spikes in the non-lead inputs that both companies rely on — not a company-specific issue, but an industry-wide cost shock.





PART THREE

The Indian Battery Industry

A mature, consolidated lead-acid duopoly sits on top of an early-stage, capital-intensive race to localise lithium-ion.

Two Markets, Two Very Different Stages of Maturity

Lead-Acid

Mature · consolidated · duopoly

- Exide and Amara Raja together dominate the organized market
- Covers automotive (OEM + replacement) and industrial applications
- Smaller organized players + a meaningful unorganized/informal recycling sector alongside them
- Growth now driven by vehicle-parc expansion and replacement cycles, not technology change

Lithium-Ion

Early-stage · capital-intensive · multi-player race

- Almost entirely import-dependent today (cells and upstream materials, overwhelmingly from China)
- Operational domestic capacity: under 1 GWh to low single-digit GWh
- Announced pipeline: 180–220+ GWh across all projects
- That gap — ambition vs. delivered capacity — is the central execution risk for the whole sector

Both Exide and Amara Raja sit at the intersection: incumbent leaders in chain one, using its cash to fund entry into chain two — where they compete against better-capitalised conglomerates (Reliance, Tata) and EV-first entrants (Ola Electric).

What's Driving Growth — and Government's Role

GROWTH DRIVERS

Auto parc & replacement cycle

Underpins steady, mid-to-low-double-digit lead-acid volume growth

Electrification of mobility

2W/3W/4W EV adoption is the single largest driver of new lithium-ion capacity

Renewables & grid balancing

Solar/wind build-out drives BESS demand via SECI, NTPC, Adani Green, JSW, Tata Power tenders

Data centre & telecom growth

Sustains lead-acid UPS demand while also being a fast lithium-ion conversion front

GOVERNMENT POLICY

GST reform

Battery GST cut 28%→18%; solar equipment 12%→5% — a direct, dated demand catalyst both companies credit for H2 FY26 strength

PLI for Advanced Chemistry Cells

Principal central incentive for domestic Li-ion capacity; Reliance and Ola Electric are cited PLI allocation recipients

Import duty structure

Up to ~18% on finished cells/batteries — a real cost disadvantage for imports once domestic capacity is price-competitive

Critical minerals strategy

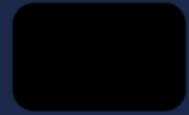
KABIL (NALCO/Hindustan Copper/MECL JV) tasked with securing overseas lithium/cobalt/nickel mining rights

India’s Lithium-Ion Gigafactory Pipeline (Mid-2026)

Initiator	Target Capacity	Chemistry Focus	Commercialisation
Reliance Industries	~120 GWh ultimate	Heavily LFP + sodium-ion (Faradion)	Ramping through H2 2026+
Tata (Agratas)	~20 GWh Phase 1	Balanced NMC/LFP	2027
Ola Electric	~20 GWh ultimate	Primarily NMC (cylindrical)	Operational, expanding
Exide (EESL)	12 GWh ultimate	50:50 NMC/LFP split	Customer samples from May 2026
Amara Raja (ARACT)	16 GWh ultimate	NMC first, scaling to LFP (BESS)	Giga-1 SOP targeted H1 CY2027

Highlighted rows: the two companies at the centre of this presentation.

Announced capacity (180–220+ GWh) may substantially exceed near-term demand (115–210 GWh/yr projected by 2030–31) — a real overcapacity/pricing-pressure risk to monitor in the back half of this decade.



PART FOUR

Exide Industries Limited

India's largest lead-acid battery maker, 79 years old, debt-free — now building a 12 GWh lithium-ion gigafactory in Bengaluru.

India's Largest Lead-Acid Battery Manufacturer

79 yrs

in continuous operation, listed since 1979

13

manufacturing plants (11 core + 2 EESL)

70+

countries served; 1,20,000+ network touchpoints

₹30,647 cr

market cap (30 Apr 2026)

OWNERSHIP & GOVERNANCE

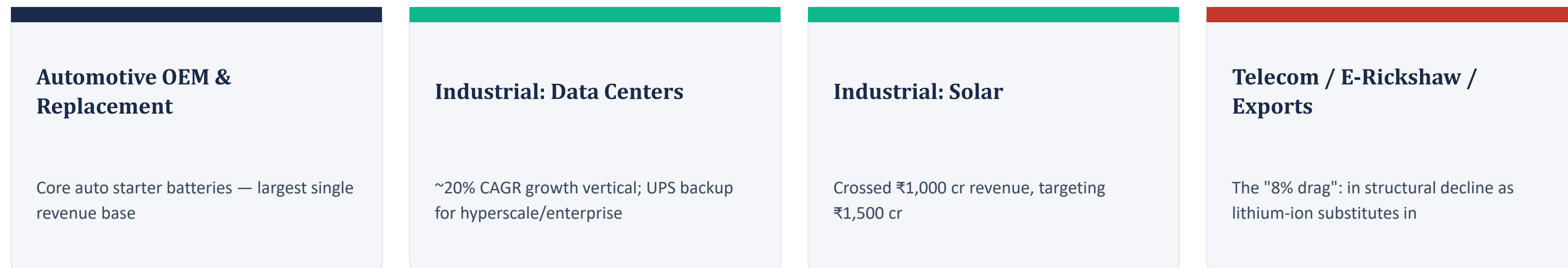
- Part of the Rajan Raheja Group since 1993; promoters hold ~46% of equity
- AAA/A1+ credit rating (ICRA); standalone balance sheet debt-free for over a decade
- Profitable every year since inception; ~16% average dividend payout ratio
- Avik Roy (MD & CEO) leads; Pravin Saraf appointed MD & CEO of new-energy subsidiary EESL (Sep 2025)

HISTORY IN BRIEF

- 1946: Manufacturing begins · 1979: Listed on exchanges
- 1993: Acquired by Rajan Raheja Group
- 2018: Nexcharge JV (Li-ion packs) with Leclanché
- 2022: EESL incorporated — dedicated Li-ion cell subsidiary
- 2025–26: Bengaluru gigafactory nears commissioning

One Exide: Trade, Institutional & International

Since FY25, Exide runs a functional model across three clusters rather than the older Automotive/Industrial split. ~92% of standalone revenue sits in the growing "core"; the balance (Telecom, E-Rickshaw, Exports) is in structural decline.



Why this matters: consolidated revenue grew only 2.79% (FY25) and 4.39% (FY26) — much slower than the "92% growing" core alone would suggest. This raises a genuine question of whether lithium-ion migration is eating into the core's most profitable growth segments faster than the core can organically replace them.

Exide Energy Solutions (EESL)

Capex vs. FY26 Actual (₹ cr)



Two facilities

Prantij, Gujarat (1.5 GWh) — operating pack/module assembly, ex-Nexcharge. Bengaluru gigafactory — Phase 1: 6 GWh of an eventual 12 GWh, split 3 GWh NMC cylindrical + 3 GWh LFP prismatic. Technology partner: SVOLT Energy (China).

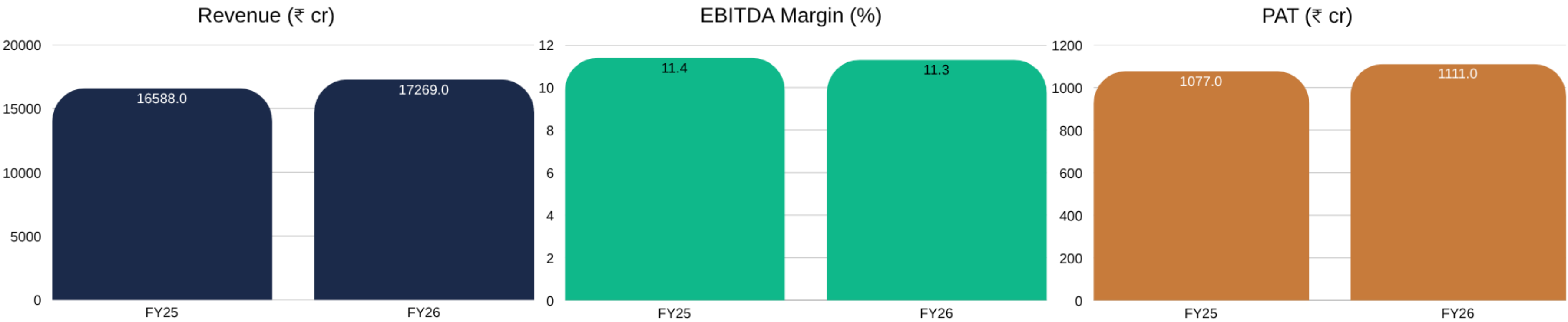
Timeline to revenue

Cylindrical (NMC) customer samples from May 2026 → Prismatic (LFP) samples June–July 2026 → Commercial revenue guided from ~Q3 FY27. Separate, incremental Hyundai/Kia passenger-EV supply agreement requires additional capex beyond the ~₹6,000 cr already committed.



Key targets to watch: >85% utilisation and 90% yield once production begins. Commercialisation has already slipped versus the original 27–30 month window (2022 announcement), and Phase 1 cost has crept up ~7.7% (₹5,200 cr → ₹5,600 cr) before first commercial shipment.

Exide Standalone: A Steadily Improving Cash Engine



Core ROCE

26.0% → 28.8%

FY25 → FY26

Asset turnover

3.1x → 3.4x

FY25 → FY26

Operating cash flow

₹1,298 cr → ₹2,231 cr

FY25 → FY26

HDFC Life stake

~₹5,000–5,140 cr

Funding cushion for EESL capex

Note: core-business ROCE excludes HDFC Life stake and EESL investment. A separate, lower disclosed series (16.3–17.6%) uses a different capital base — flagged in the dossier as unreconciled.

Exide: Bull Case vs. Bear Case

BULL CASE

- Core lead-acid ROCE rising (26.0%→28.8%) while funding Li-ion entry without diluting shareholders
- "Last man standing" dynamic as global lead-acid competitors retreat
- Multiple megatrend verticals: Data Centers (~20% CAGR), Solar, rail electrification
- 25-year share price CAGR of 19.0% — ahead of Nifty 50 (12.6%) and Sensex (12.7%)
- HDFC Life stake (~₹5,000+ cr) is a funding cushion for the EESL capex programme

BEAR CASE

- Consolidated growth (2.79%/4.39%) suggests Li-ion migration may be eating the core's best growth segments
- Bengaluru gigafactory commercialisation has already slipped; no firm dispatch date as of May 2026
- Phase 1 cost already up ~7.7% before first commercial cell shipped
- HDFC Life stake is volatile/mark-to-market (~₹714–850 cr negative OCI hit in FY26) — a treasury mismatch if monetised under pressure
- Hyundai/Kia relationship requires a second, incremental capex cycle atop an unproven first one

Every Revenue Segment, End to End

Automotive • OEM: 2W / 3W / 4W new-vehicle supply (Institutional) • Replacement / Aftermarket (Trade) — highest-margin, most defensible line	Industrial (B2B) • Industrial UPS incl. Data Centers • Railways / Traction / Motive Power • Telecom (structurally declining) • Power & Projects / Industrial Infra • Solar — Institutional (grid-scale utility)	Consumer / Trade • Home UPS / Inverter batteries (seasonal) • Solar — Trade (off-grid battery+panel+inverter)
International & Specialty • Exports (≈70+ countries) • Defence & Submarine Batteries (since 1992, niche)	New Energy (EESL) • Li-ion cells: NMC cylindrical + LFP prismatic • Packs & modules (Nexcharge, Prantij) • Stationary storage / BESS-oriented LFP line	 11 distinct segments, one balance sheet Auto OEM+Aftermarket ≈ 50% of revenue; Industrial ≈28–32%; Exports ≈5%; New Energy reported separately via EESL.

Revenue Growth Assumptions — Bear / Base / Bull

Source: Exide Standalone Model, Assumptions tab. Growth rates applied to FY26A revenue base of ₹17,268.92 cr (95% Domestic / 5% Export; Domestic split 25% Institutional / 75% Non-Institutional).

Domestic Institutional Revenue Growth %

	FY27E	FY28E	FY29E	FY30E	FY31E
Bear	6.0%	5.5%	5.0%	5.0%	5.0%
Base	8.0%	7.5%	7.0%	6.5%	6.0%
Bull	9.5%	9.0%	8.0%	7.0%	6.5%

Domestic Non-Institutional Revenue Growth %

	FY27E	FY28E	FY29E	FY30E	FY31E
Bear	10.0%	9.5%	9.0%	8.5%	8.0%
Base	13.0%	12.0%	11.0%	10.0%	9.5%
Bull	13.5%	13.0%	12.0%	11.0%	10.0%

Export Revenue Growth % (FY27E explicit)

	FY27E	FY28E	FY29E	FY30E	FY31E
Bear	−8.0%	—	—	—	—
Base	−4.0%	—	—	—	—
Bull	0.0%	—	—	—	—

From FY28E onward, Exports are modelled to recover toward a target of 8% of total standalone revenue (all scenarios) — a single-year jump the model itself flags as aggressive.

Resulting Total Revenue (₹ cr)

Bear (FY31E)	₹26,346.6 cr
Base (FY31E)	₹28,882.1 cr
Bull (FY31E)	₹30,005.3 cr
FY26A actual base	₹17,268.9 cr

MODEL ASSUMPTIONS

Cost, Capex & Discount Rate Assumptions

COST STRUCTURE (scenario-invariant)

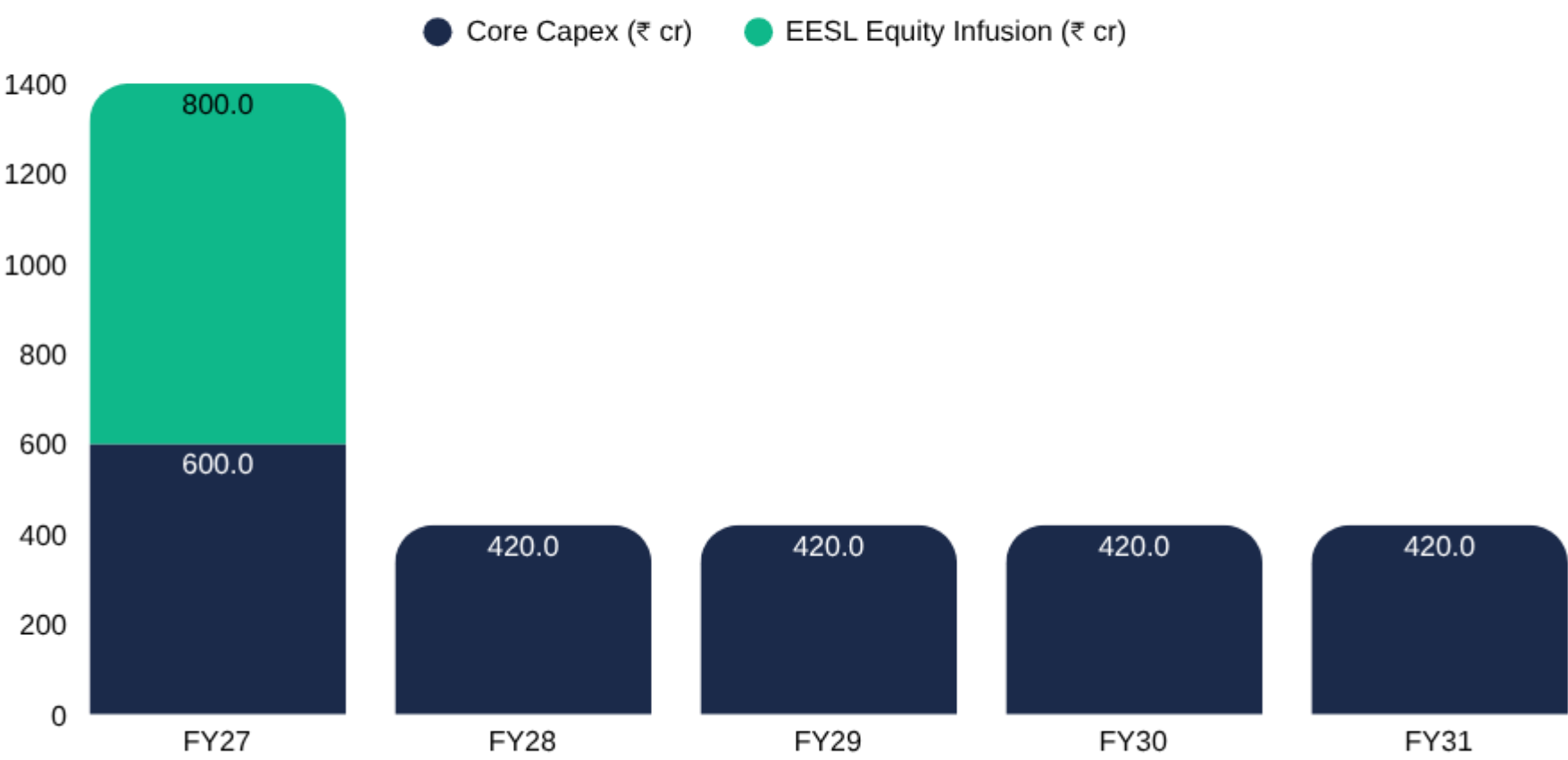
COGS	70.0% of revenue
Other Expenses — Fixed	₹1,590.76 cr base (9.2% of FY26 rev.), +8%/yr
Other Expenses — Variable	10.1% of revenue
Depreciation rate	16.6% of opening net block
Tax rate	25.0%
Dividend	₹170 cr/yr, flat
Working capital days	Inventory 94.1 · Receivable 32.2 · Payable 92.6 (held at FY26A)

DISCOUNT RATE & TERMINAL VALUE

Cost of Equity (Ke, CAPM)	14.06% = Rf 6.77% + β 1.03 × ERP 7.08%
Terminal growth rate	5.0%

Note: an Executive Summary comment elsewhere in the model refers to a "12% placeholder" discount rate — the formulas themselves consistently use the 14.06% CAPM figure above (verified against the discount-factor cells).

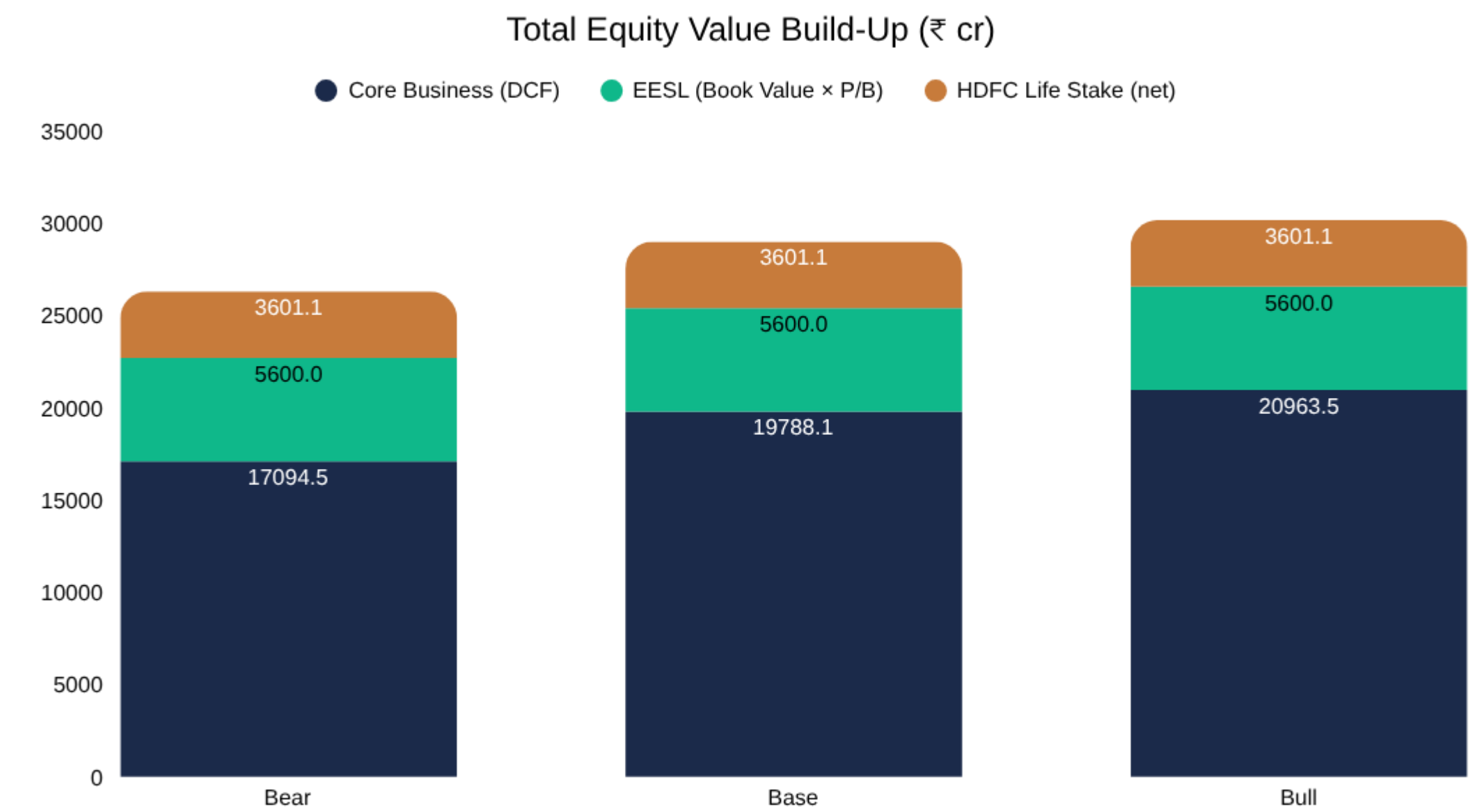
Capex Distribution Across Forecast Years (₹ cr)



EESL cumulative investment

₹4,802.2 cr infused by FY26A-end → ₹5,602.2 cr cumulative by FY27E (final ₹800 cr tranche) → held flat FY28E–FY31E. This ₹5,600 cr book value is the basis for the EESL component in the SOTP (next slide).

Exide — Sum-of-the-Parts by Scenario



Value Per Share (Rs.)

Bear	₹309.36
Base	₹341.05
Bull	₹354.88

Total Equity Value (Rs. cr)

Bear ₹26,295.6 · Base ₹28,989.2 · Bull ₹30,164.6

Shares outstanding: 85 cr. EESL and HDFC Life components are held constant across all three scenarios — only the core DCF varies with revenue growth.

KEY RATIOS

Exide Standalone — FY24A to FY31E

Metric	FY24A	FY25A	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E
Revenue Growth %	n/a	3.5%	4.1%	11.9%	16.3%	11.1%	10.1%	9.2%
EBITDA Margin %	11.7%	11.4%	11.3%	11.3%	12.1%	12.5%	12.9%	13.1%
PAT (₹ cr)	1,053	1,077	1,111	1,210	1,631	1,998	2,380	2,772
PAT Margin %	6.6%	6.5%	6.4%	6.3%	7.3%	8.0%	8.7%	9.2%
ROCE (Core) %	25.1%	24.4%	29.4%	31.9%	33.2%	31.1%	28.6%	26.2%
ROE %	8.0%	7.5%	7.6%	7.7%	9.5%	10.5%	11.2%	11.6%
P/E (x)	33.8x	33.1x	32.1x	29.4x	21.8x	17.8x	15.0x	12.9x
EV/EBITDA (x)	19.1x	19.0x	18.4x	16.4x	12.6x	10.4x	8.6x	7.1x

Source: Exide Standalone Model, Ratios tab (Base scenario forecast). ROCE shown on core basis, excluding EESL and HDFC Life. Market multiples (P/E, EV/EBITDA) anchor to the model’s reference CMP of ₹419.20/share (₹35,632 cr market cap ÷ 85 cr shares).



PART FIVE

Amara Raja Energy & Mobility

The other half of India's lead-acid duopoly — pursuing a more diversified, four-entity push into cells, recycling, chargers and BESS.

The Other Half of the Lead-Acid Duopoly

40+ yrs

founded by Dr. Ramachandra N Galla

4

distinct new-energy entities: ARACT, ARCS, ARPS, ARBME

₹13,814 cr

FY26 consolidated revenue

0.00–0.03x

gearing — historically debt-free at group level

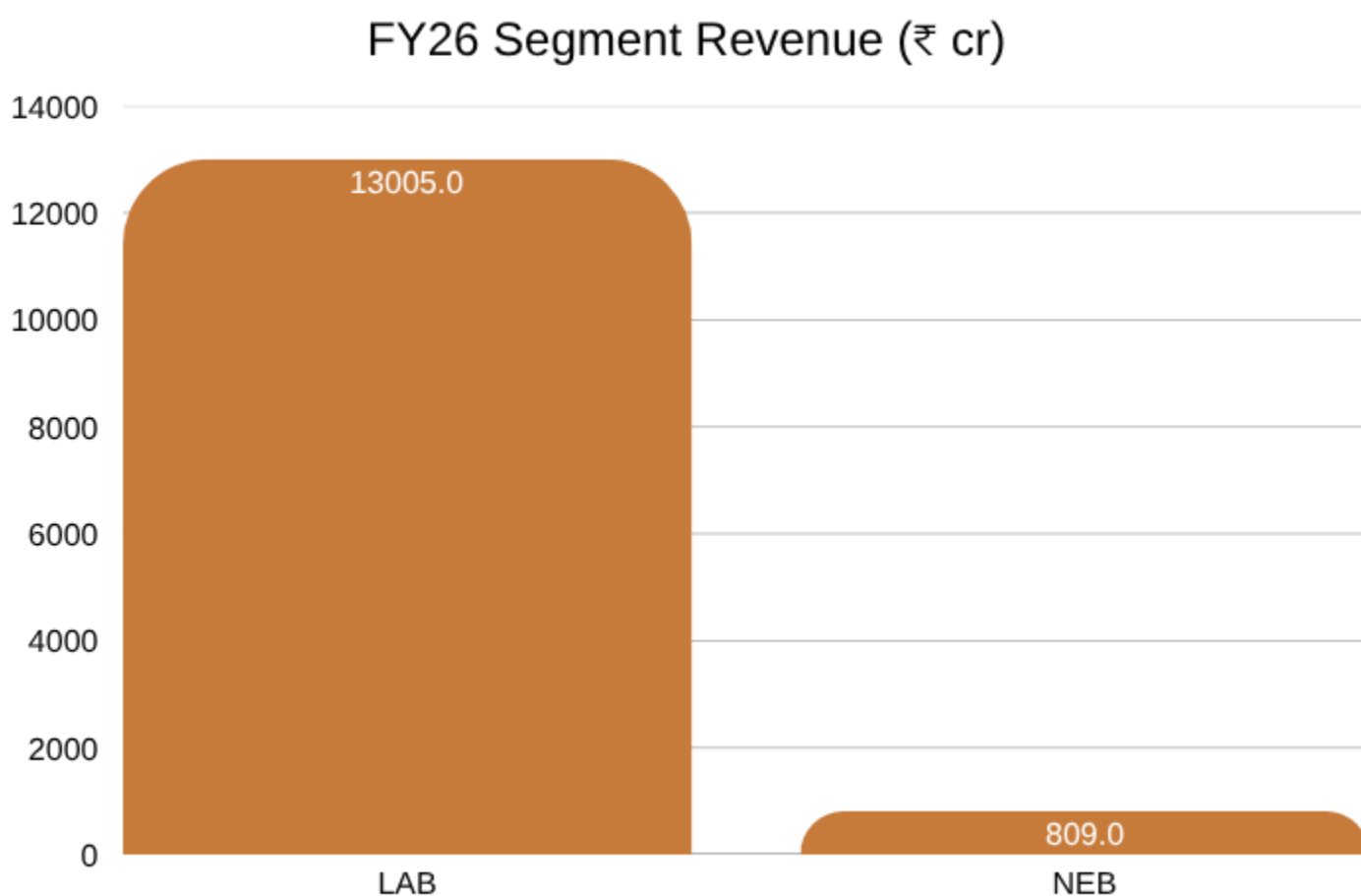
OWNERSHIP & STRATEGY

- Chairman & MD: Jayadev Galla; Galla family promoter holding ~28.06%
- Renamed from "Amara Raja Batteries" to "Energy & Mobility" in 2023 to reflect the pivot
- 21-year Johnson Controls JV (1995–2019) built the VRLA/AMARON technology base before ending involuntarily on JCI’s own global divestment
- Explicit "dual-pillar" strategy: LAB cash funds the New Energy Business (NEB) build-out

FOUR NEW-ENERGY ENTITIES

- ARACT — Li-ion cells & Giga Corridor (16 GWh target)
- ARCS — lead recycling (Cheyyar, Tamil Nadu)
- ARPS — chargers & power electronics (5,000+ unit order book)
- ARBME — Middle East/Africa marketing arm

Lead-Acid Batteries (LAB) + New Energy Business (NEB)



LAB: growing, high-margin core

Automotive + Industrial verticals. FY26 revenue ~₹13,005 cr, +5–8% YoY. Distribution moat: 100,000+ touchpoints, 3,000+ franchisees in the aftermarket.

NEB: fast-growing, still loss-making

Revenue surged 61.4% YoY (FY25→FY26) off a small base — but posted a segment loss of ~₹135 cr (~ -16.7% EBIT margin). ARACT alone: loss widened 6x YoY (₹12.14 cr → ₹73.23 cr) even as revenue grew 61%.



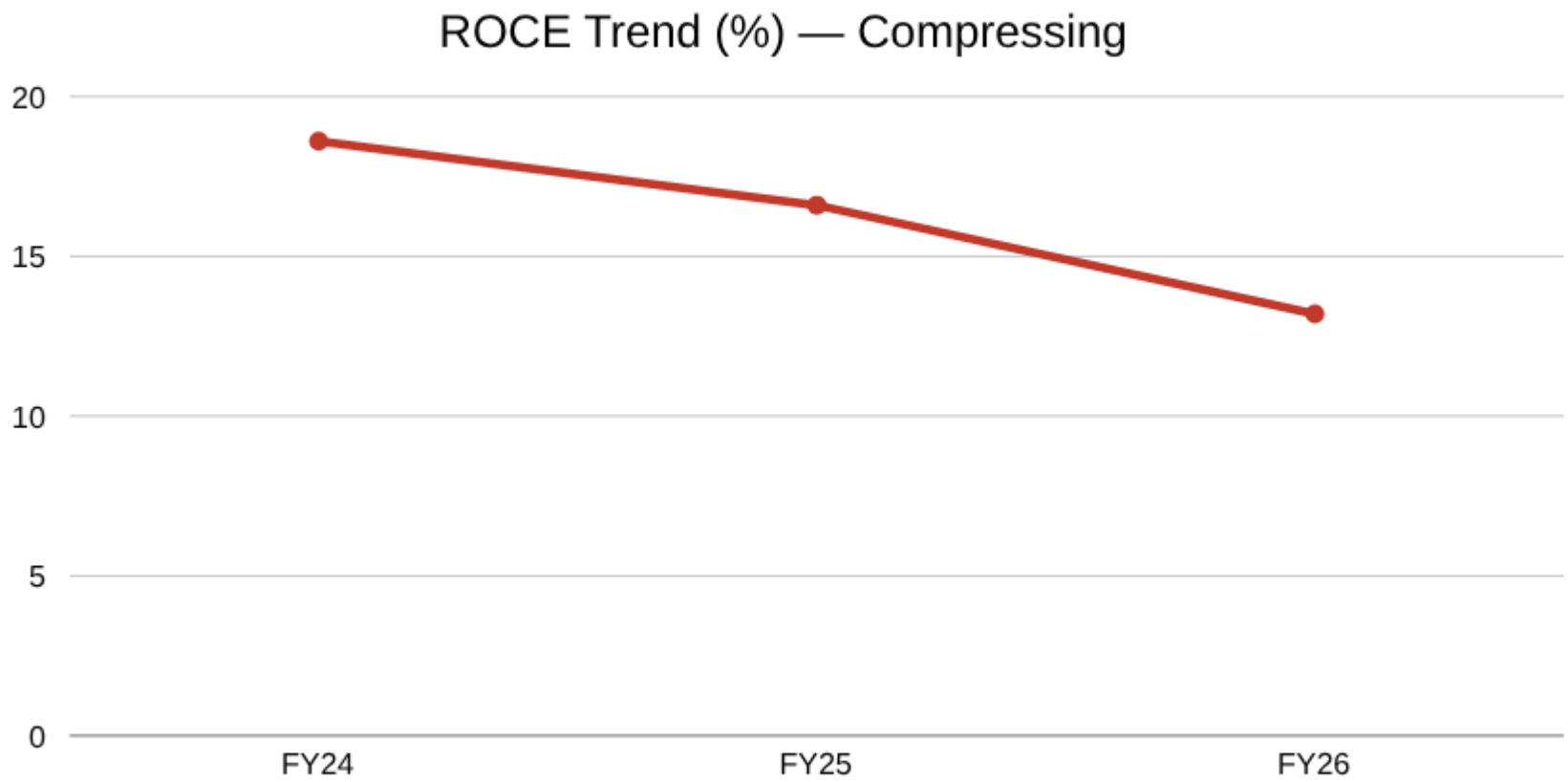
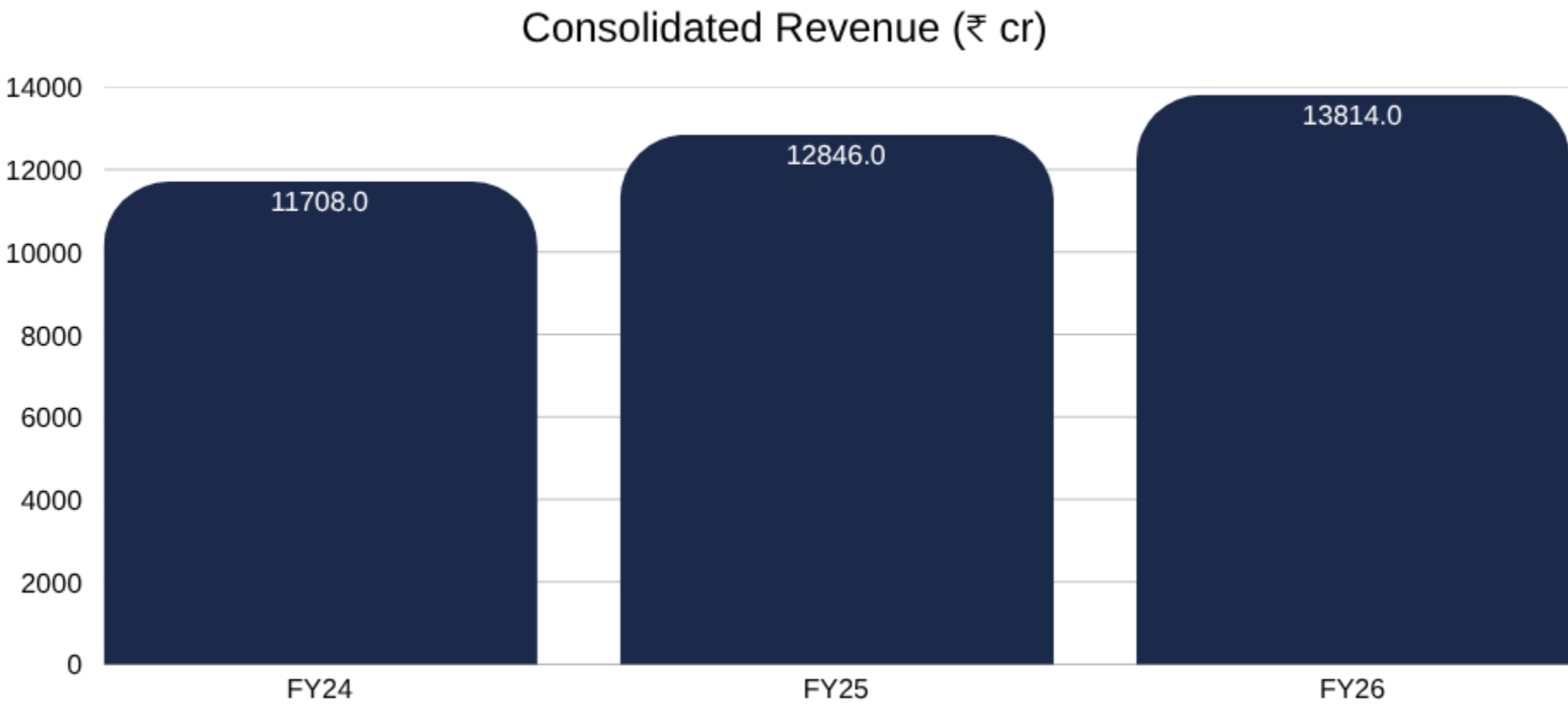
Track record flag: the FY25 NEB revenue guidance of ₹1,000 cr was missed by roughly 50%. This should discount the credibility of the current FY27 target (7–8% revenue share, implying ~50–60% NEB growth) until proven.

The Amara Raja Giga Corridor

Giga-1 (Cylindrical)	2 GWh Phase 1 of a 16 GWh ultimate target · NMC cylindrical cells · technology partner Gotion High-Tech (via GIB EnergyX) · targeting H1 CY2027 start of production
BESS Facility	Separate 5 GWh facility · targeted Q4 FY27/end-CY2026 · LFP-leaning chemistry · ~₹280 crore capex drawdown disclosed
Customer Qualification Plant	Multi-chemistry pilot line · commissioned in FY26 · targeting full-scale operation and 4W-OEM sample qualification by Q2 FY27
Total commitment	~₹9,500 crore Giga Corridor capex · modelled at ~70%-funded by debt by FY31 — a materially more leveraged posture than Exide’s

Offtake caveat: Ather Energy and Piaggio relationships are MoUs for "long-term product alignment" — not confirmed multi-year supply contracts. Demand risk for the bulk of committed capacity remains genuine and unconfirmed.

Amara Raja Consolidated: Growing, but Returns Compressing



FY26 PAT

₹896 crore

consolidated, LAB + NEB combined

EBITDA margin

10.8%

below mgmt’s own 14% long-term aspiration

Material cost/sales

68.5% → 69.7%

FY25 → FY26

Lead price base

₹1.5 lakh/t → ₹2.1 lakh/t

structural re-basing over 3 years

Amara Raja: Bull Case vs. Bear Case

BULL CASE

- Genuine distribution moat in high-margin aftermarket (100,000+ touchpoints, 3,000+ franchisees)
- ARPS charger business is commercially live — India's first IP67-localised off-board charger, 5,000+ unit order book
- "Chemistry-agnostic" strategy retains Telecom/export customers through the transition; 1 GWh+ telecom lithium-pack installations already live
- Conservative historical balance sheet (gearing 0.00–0.03x) funding a record capex programme
- BESS facility offers a potentially faster path to EBITDA-accretive scale (9–10x claimed asset turnover) than cell manufacturing

BEAR CASE

- Demonstrated track record of missing NEB targets (~50% miss on FY25's ₹1,000 cr guidance)
- EBITDA margin (10.8%) well below management's own 14% aspiration, with a multi-year path back
- ROCE already declined 19.2%→16.2% (FY24→FY25); FY27 capex guided even higher
- Management concedes it cannot beat Chinese cell import prices today — targeting only "China plus 15–20%"
- Ather/Piaggio offtake are MoUs, not binding contracts; four-entity execution spreads management bandwidth thin

Every Revenue Segment, End to End

Automotive & Adjacent

- Amaron® (premium 4W/2W, AGM/EFB start-stop)
- PowerZone™ (value/rural) · ELITO® (international)
- 3W, commercial-vehicle & farm equipment
- Home UPS/Inverter · Lubricants (Amaron Hi-Life)

Industrial (B2B)

- Telecom (leadership; LAB share now <5%)
- UPS & Data Centers (largest installed base claim)
- Railways (AC/lighting, integrated power)
- Power & Utilities, Oil & Gas, Solar integrators
- Motive Power (aviation cargo, forklift fleets)

New Energy Business

- ARACT — Li-ion cells, Giga Corridor (16 GWh)
- BESS — stationary grid/telecom storage (5 GWh)
- ARPS — chargers & power electronics
- ARCS — lead recycling (Cheyyar, TN)
- ARBME — Middle East / Africa export arm

17 segments across 3 categories — Automotive/Adjacent (≈67–71% of revenue), Industrial (≈25–29%), and a fast-growing but still loss-making New Energy Business (≈6% of FY26 revenue).

LAB Revenue Growth Assumptions — Bear / Base / Bull

Source: ARE&M Model, Assumptions tab. Applied to FY26A base: Replacement ₹6,463.6 cr · OEM ₹2,770.1 cr · Industrial ₹3,771.6 cr (LAB total ₹13,005.3 cr).

Replacement (Aftermarket) Revenue Growth %

	FY27E	FY28E	FY29E	FY30E	FY31E	FY32E
Bear	9.0%	8.5%	9.0%	8.5%	8.0%	8.0%
Base	12.0%	11.5%	12.0%	11.5%	11.0%	10.0%
Bull	15.0%	14.0%	14.0%	13.0%	12.0%	11.0%

OEM Revenue Growth %

	FY27E	FY28E	FY29E	FY30E	FY31E	FY32E
Bear	6.0%	6.0%	6.0%	6.0%	6.0%	6.0%
Base	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%
Bull	10.0%	9.0%	9.0%	9.0%	9.0%	9.0%

Industrial Revenue Growth %

	FY27E	FY28E	FY29E	FY30E	FY31E	FY32E
Bear	5.0%	5.5%	6.0%	6.5%	7.0%	7.5%
Base	7.0%	7.5%	8.0%	8.5%	8.0%	8.0%
Bull	9.0%	9.5%	10.0%	10.5%	11.0%	11.5%

Resulting LAB Revenue (FY32E): Bear ₹19,899.3 cr · Base ₹22,633.0 cr · Bull ₹25,031.1 cr. Industrial growth is deliberately modelled to RISE across the forecast as the Telecom drag (already <5% of LAB revenue) shrinks while Data Center/UPS demand (AI buildout, data localisation) continues.

MODEL ASSUMPTIONS

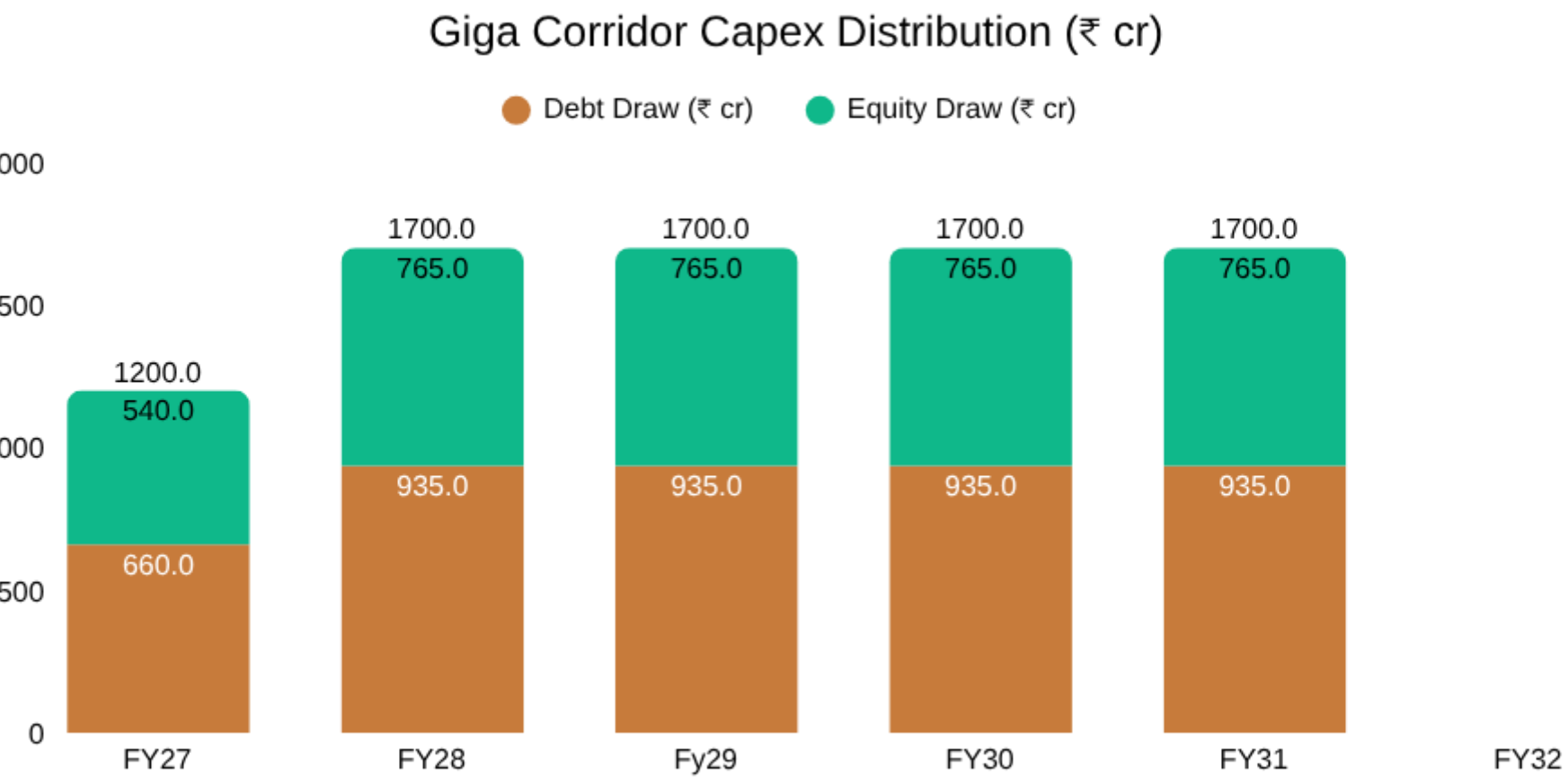
NEB Growth, Capex & Discount Rate Assumptions

NEB — Lithium Packs Growth %

	FY27E	FY28E	FY29E	FY30E	FY31E	FY32E
Bear	20%	18%	16%	15%	15%	15%
Base	25%	22%	20%	18%	18%	18%
Bull	30%	25%	22%	20%	18%	16%

NEB — Chargers & Power Systems Growth %

	FY27E	FY28E	FY29E	FY30E	FY31E	FY32E
Bear	10%	9%	8%	8%	8%	8%
Base	15%	14.5%	14%	13.5%	13%	12.5%
Bull	18%	17%	16%	15%	14%	13%

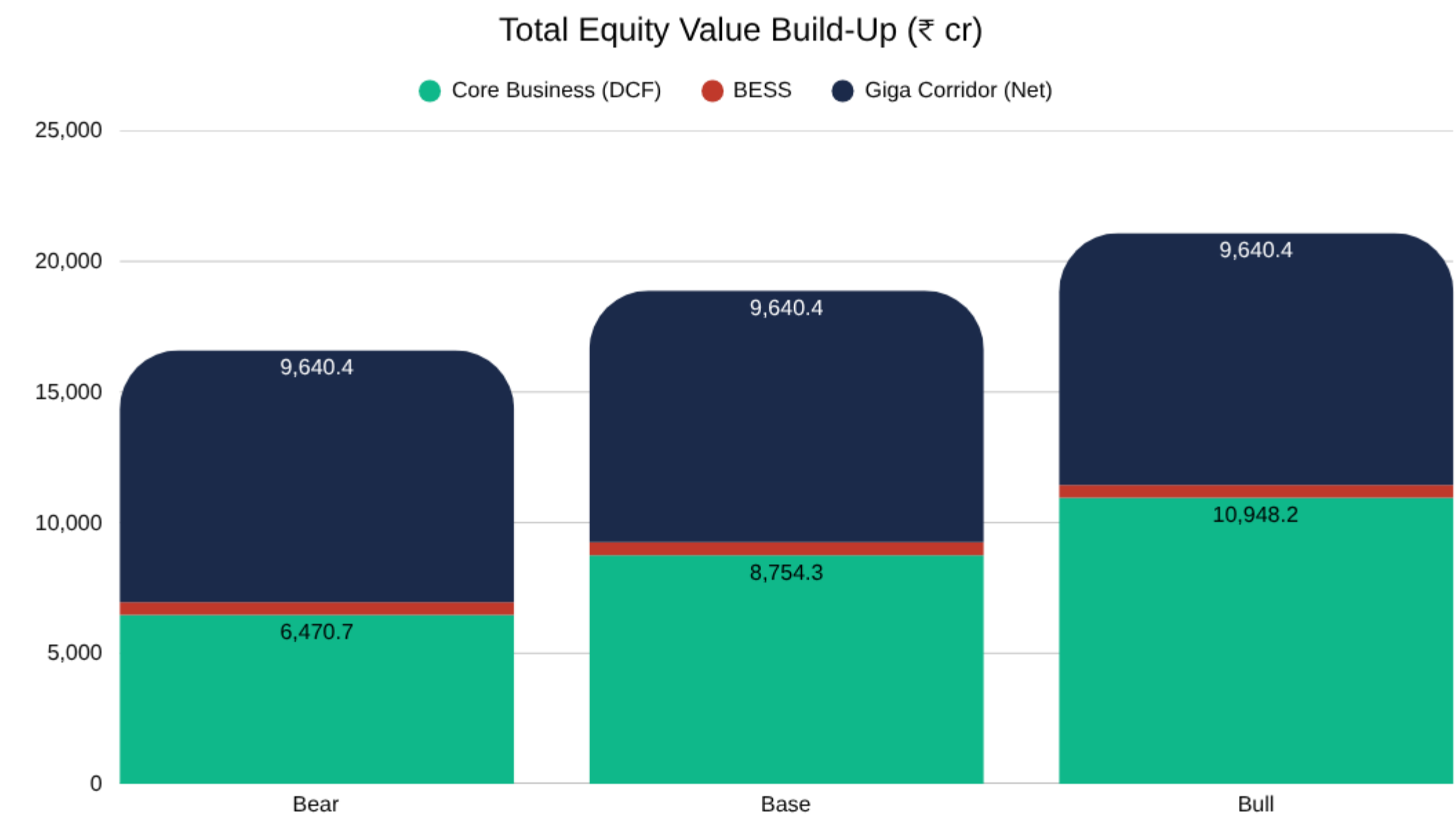


Cost, WC & Discount Rate	
LAB COGS	70.0% of LAB revenue
Employee cost (FY26 base)	₹839.6 cr, +6%/yr
Other Exp. — Fixed / Variable	₹293.7 cr @ +6%/yr / 11.0%→10.5% of rev.
WC days (Inv/Recv/Pay)	84.3 / 31.3 / 42.7
Tax rate · Dividend	25.0% · ₹194 cr/yr flat (≈₹10.60/sh)
Cost of Equity (Ke, CAPM)	13.85% = Rf 6.77% + β1.0×ERP7.08%
Terminal growth rate	5.0% (sensitivity: 4%–6%)

₹9,500 cr total Giga Corridor commitment

₹1,500 cr already invested to Mar-26; ₹8,000 cr remaining (₹1,200 cr FY27E, ₹1,700 cr/yr FY28E–FY31E). Financed via the schedule’s actual computed 55:45 debt:equity split (labelled "70%" in the source file — transcribed as-is). Cumulative debt reaches ₹4,400 cr by FY31E, held flat in FY32E.

Amara Raja — Sum-of-the-Parts by Scenario



Value Per Share (Rs.)

Bear	₹907.1
Base	₹1,031.9
Bull	₹1,151.8

Total Equity Value (Rs. cr)

Bear ₹16,603.0 · Base ₹18,886.6 · Bull ₹21,080.5

CMP (31-Mar-26): ₹876/share. Implied upside/(downside): Bear 3.6%, Base 17.8%, Bull ~31.5% — the Bull case lands almost exactly at CMP. Shares outstanding: 18.30 cr.

KEY RATIOS

Amara Raja Consolidated — FY24A to FY32E

Metric	FY24A	FY25A	FY26A	FY27E	FY28E	FY29E	FY30E	FY31E	FY32E
Revenue Growth %	n/a	9.7%	7.5%	10.5%	10.3%	10.7%	10.5%	10.2%	9.8%
EBITDA Margin %	14.2%	12.6%	10.8%	9.4%	10.2%	11.1%	11.6%	12.1%	12.4%
PAT (₹ cr)	934	945	896	765	950	1,164	1,350	1,557	1,750
PAT Margin %	8.0%	7.4%	6.5%	5.0%	5.6%	6.2%	6.5%	6.9%	7.0%
ROCE %	17.3%	15.4%	11.5%	11.0%	12.5%	13.8%	14.2%	14.6%	15.0%
ROE %	13.7%	12.8%	11.1%	8.8%	10.1%	11.2%	11.7%	12.1%	12.1%
P/E (x)	17.2x	17.0x	17.9x	21.0x	16.9x	13.8x	11.9x	10.3x	9.2x
EV/EBITDA (x)	9.7x	9.9x	10.7x	11.2x	9.4x	7.7x	6.7x	5.8x	5.2x

Source: ARE&M Model, Ratios tab (Base scenario forecast; historicals are reported consolidated actuals). Market multiples (P/E, EV/EBITDA) anchor to CMP of ₹876/share (31-Mar-26).



PART SIX

Head-to-Head & Valuation

Putting Exide and Amara Raja side by side — financials, capital structure, valuation, and the risks that matter most.

Exide vs. Amara Raja — FY26 Side by Side

Metric	Exide (Standalone)	Amara Raja (Consolidated)
Revenue	₹17,269 cr (+4.1% YoY)	₹13,814 cr (+7.53% YoY)
PAT	₹1,111 cr (6.4% margin)	₹896 cr
Core ROCE	26.0% → 28.8% (rising)	16.6% → 13.2% (declining)
Balance sheet	Debt-free (standalone); AAA/A1+	Debt-free historically; ~70%-debt-funded Giga Corridor modelled
New-energy structure	Single subsidiary (EESL)	Four entities (ARACT, ARCS, ARPS, BESS)
Ultimate Li-ion capacity	12 GWh	16 GWh
Distinct embedded asset	HDFC Life stake (~₹5,000+ cr)	None equivalent

On disclosed figures, Exide’s core business shows a stronger, more stable return profile; Amara Raja’s is more diluted by New Energy losses and a more debt-funded build-out — though Exide’s own ROCE methodology carries an unresolved internal discrepancy (16–18% vs. 26–29% series).

Where Each Company Has the Advantage

Exide's Advantage

- Debt-free, higher-disclosed-return core business
- Simpler, single-subsidiary new-energy structure (easier to monitor)
- HDFC Life stake — a distinct embedded-value pool with no Amara Raja equivalent
- More advanced primary-source verification in its own research base

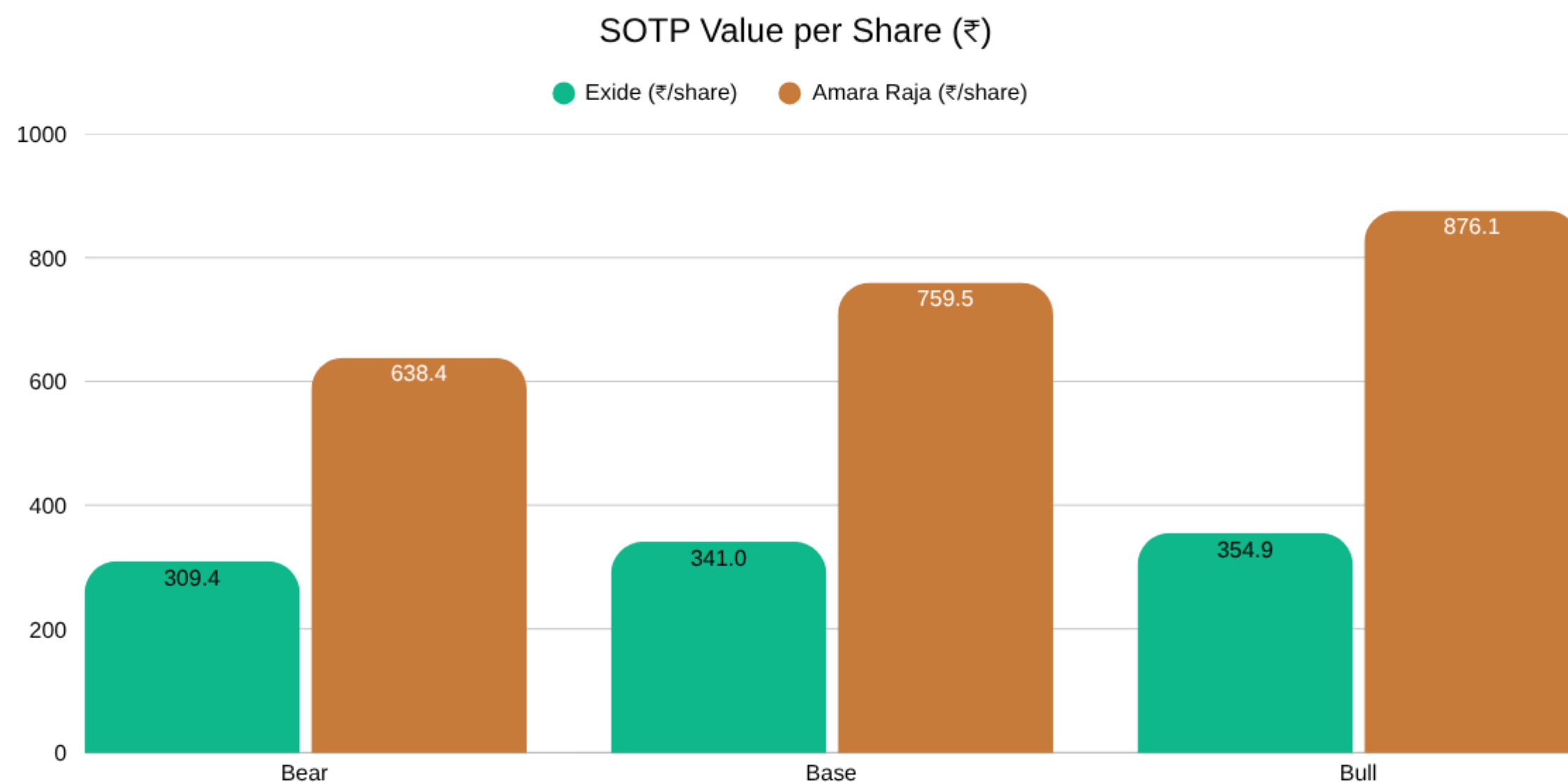
Amara Raja's Advantage

- More diversified new-energy structure — four separately trackable vectors
- Early, already-revenue-generating lead in distributed lithium BESS (telecom-site storage)
- Explicit FY27 LAB growth guidance (8–10%) gives a clearer near-term anchor
- Live, commercially operating charger business (ARPS) as a proof point

Both companies independently arrive at the same mental model: the lead-acid business as a "synthetic bond" funding a "new-energy call option." That two separate research threads converge on this framing strengthens confidence it reflects a genuine industry-structural pattern.

Sum-of-the-Parts: Bear / Base / Bull

Both models value the core lead-acid business on DCF, and the new-energy subsidiaries at Book Value × an assumed Price-to-Book multiple — deliberately not borrowing the company’s own trading multiple, which would be circular.



Reading the numbers

- Amara Raja CMP referenced in the model: ₹876 — Base case implies (13.3)% downside; even the Bull case lands almost exactly at CMP
- Exide’s model does not yet carry a live CMP in its Executive Summary — flagged as a gap to close
- Both SOTPs use a placeholder discount rate (Exide: 12%) pending a formally derived WACC

What the Market May Be Missing

The market narrative on both stocks tends to oscillate between two extremes: "legacy compounder facing terminal EV disruption" or "speculative Li-ion story stock." The more useful framing keeps the two businesses analytically separate.

The Cash Machine

A highly efficient, high-ROCE, self-funding legacy lead-acid business — the "synthetic bond" half of the story, real and improving on the numbers disclosed.

The Call Option

A high-risk, high-optionality industrial start-up (Li-ion cells) housed in the same corporate shell — unproven, capital-intensive, and years from a clean return profile.



The central valuation question: for both stocks, notably, even the model's Bull case for Amara Raja lands almost exactly at today's market price — implying the market may already be pricing in an outcome at or beyond this model's most optimistic new-energy scenario. Whether that's justified is the question worth revisiting every quarter.

Common and Company-Specific Risks

COMMON TO BOTH COMPANIES

- Commodity-cost volatility — lead, antimony, tin, silver, sulfur (core); lithium, nickel, cobalt (new energy)
- Gigafactory execution risk — yield/utilisation ramp-up is unproven at scale for either company
- Single foreign technology-transfer partner exposure (SVOLT for Exide, Gotion for Amara Raja) to Chinese export-control policy
- Guidance-credibility risk — both dossiers independently flag a pattern of favourable management framing

Exide-Specific

- Core-ROCE and inventory-days disclosure inconsistencies (data-discipline risk)
- HDFC Life stake volatility as an implicit funding backstop

Amara Raja-Specific

- More leveraged capex funding posture for the Giga Corridor
- Wider set of unresolved internal revenue/growth-rate inconsistencies flagged in the dossier

A Quarterly Monitoring Checklist

Sector	Operational (not announced) Li-ion capacity growth; LFP/NMC mix; China export-control dates; BESS tender activity
Exide	EESL utilisation%/yield% once production starts; firm commercial-dispatch date; HDFC Life stake monetisation signals
Amara Raja	Giga-1 SOP progress toward H1 CY2027; CQP 4W-OEM qualification (Q2 FY27); Ather/Piaggio MoU-to-contract conversion
Both	Guidance hit/miss scorecard every quarter — the single highest-value recurring exercise given demonstrated guidance-miss patterns

Two Quality Franchises, Underwriting an Unproven Bet

- Both companies' lead-acid businesses are genuinely high-quality, cash-generative franchises — a reasonable investment case exists on the core alone
- The lithium-ion pivot is the single largest swing factor for both stocks — execution risk, not strategic logic, is the open question
- Current valuations (especially Amara Raja's, where even the Bull case sits at CMP) suggest the market may already be paying for new-energy success that has not yet been demonstrated
- Recommendation: track the quarterly monitoring checklist closely before taking a directional position — the next 2–3 quarters of EESL/Giga-1 disclosures will be decisive

Thank You

Questions & Discussion

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